

Theory of demand & supply

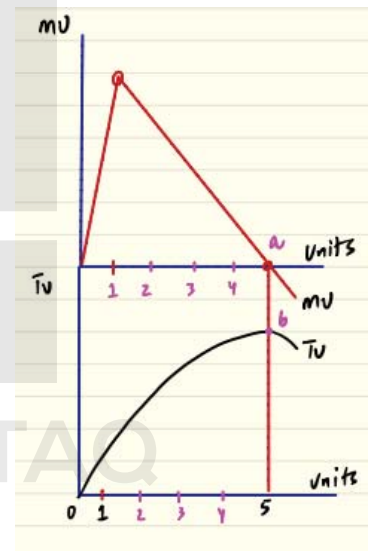
CONSUMER BEHAVIOUR CAN BE EXPLAINED USING TWO MAIN APPROACHES:

1. Marginal utility theory (the cardinal's approach)
2. Indifference curve analysis (the Ordinal's approach)

Marginal utility theory (the cardinalist approach): This approach was termed cardinal since cardinal numbers could be used to measure utility.

1. The theory was developed by Alfred Marshall who introduced an imaginary unit called the utils as a means of measuring utility.
2. This theory is based on the premise that the amount of satisfaction or utility obtained from the consumption of particular product can be measured in the same ways that physical units can be measured.
3. The consumer act rationally and attempt to get the best value for money from their purchases i.e. they tend to maximize utility.

UTILITY: The satisfaction a consumer receives from consuming a product is called utility. Total utility refers to total satisfaction from the amount of that product consumed. Marginal utility refers to the change in satisfaction resulting from consuming a little more or little less of that product. The MU of the 14th egg consumed is the additional satisfaction provided by the consumption of that unit. Thus marginal utility of the 14th egg is the difference in total utility gained from consuming 13 eggs per week and from consuming 14 eggs per week.



The basic assumption of the utility theory, sometimes called the law of diminishing marginal utility, is as follows: "the utility that any consumer derives from successive units of a particular product diminishes as total consumption of all other products remains constant"

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Theory of demand & supply

One commodity equilibrium: If we hold the consumption of other products constant, we can study the consumer's behavior and derive the demand curve from the marginal utility schedule. Let the consumer be faced with the given market price of some product. The price shows what he must sacrifice to obtain each unit of the product. The marginal utility schedule or graph shows the value he places on each unit of the product. Clearly the consumer can increase his utility by adding to his purchases as long as the value he places on each additional unit exceeds the amount he must pay for that unit. It follows that the utility maximizing consumers will adjust his purchases of product until the MU of the last unit purchased is equal to the price of that product i.e. $MU_x = P_x$. For example, at price of \$10, the consumers will buy three ice creams.

THE DEMAND CURVE: If we represent utility in monetary terms i.e. we replace MU with the price then it can be concluded that when the consumption of other products is held constant, the MU schedule for the product is that product's demand curve.

MULTI-COMMODITY EQUILIBRIUM: Assume that the consumer has a choice between two products X and Y both cost \$1 each and consumer has \$1 to spend then obviously the commodity which yields the greater utility will be bought.

If we were to apply this principle to each successive unit of the consumer's spending, we could conclude that utility will be maximized when income has been allocated in such a way that the utility derived from one extra dollar's worth of X is equal to the utility derived from the consumption of one extra dollar's worth of Y. If this calculation is not fulfilled then the consumer could obviously increase the total utility by switching the expenditures from X to Y or vice versa. For example, a consumer would not spend a dollar extra more on vegetables if it was considered that the same dollar spent on meat would give more satisfaction or in the reverse situation would not spend a dollar extra on meat if he considers that the same dollar spend on vegetables would yield greater satisfaction.

The extra satisfaction derived from the consumption of one more unit of X is its MU which we can write as MU_x and that of Y as MU_y . We must also consider the

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Theory of demand & supply

relative prices of one more unit of X and Y which we can write as P_x and P_y . We can then see that the utility maximizing condition when: $MU_x/P_x = MU_y/P_y$

Suppose that, starting from a condition of equilibrium, the price of X falls relative to Y. we now have condition where the utility from the last dollar spent on X will be greater than the utility from the last dollar spent on Y: $MU_x/P_x > MU_y/P_y$

In order to restore the equilibrium the consumer will buy more of X and less of Y, thus reducing the MU of X and increasing the MU of Y. the consumer will continue substituting X for Y until equilibrium is achieved.

Thus we have attained the normal demand relationship that, other things being equal, as the price of X falls, more of it is bought. We have therefore, a normal downward sloping demand curve i.e. the theory of diminishing MU and theory of equal MUs per \$ spent gives us a negative relation between quantity demanded of a commodity and its price, so the individual's demand curve for any good will be same as their MU curve where utility is measured in monetary terms. Market demand is the total demand of the commodity in the entire economy and the market demand curve is the horizontal summation of the individual demand curves for a particular commodity.

The idea of consumer equilibrium is shown in Table 30.2. Two products, x and y, are priced at \$1 and \$2 respectively. It is assumed that a consumer has \$10 to spend. Consumer equilibrium is where the MU/P is the same for each product. This is where 4x and 3y are consumed with a total utility score of 235. No other combination of goods generates a higher total utility; therefore, any other combination would provide less satisfaction.

Product x (\$1 each)			Product y (\$2 each)	
MU	MU/P	Quantity	MU	MU/P
60	60	1	42	21
40	40	2	32	16
25	25	3	24	12
12	12	4	18	9
5	5	5	14	7
2	2	6	12	6

Table 30.2: Consumer equilibrium

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Limitations of marginal utility theory and its assumptions of rational behavior

There are a number of limitations of marginal utility theory, particularly in relation to the law of diminishing marginal utility:

1. **Unit of measurement:** As has already been indicated, it is difficult to find an appropriate unit of measurement of utility.
2. **Habit and impulse:** Some purchases may be habit forming or made on impulse with the consumer not really considering the marginal utility.
3. **Enjoyment may increase as consumption increases:** In some cases, the amount of satisfaction gained from additional units of consumption of a product might increase as the number consumed increases. This may well be the case with collectors. A collector wishing to have the largest collection in the world of Beatles memorabilia will obtain greater satisfaction from each additional item purchased and may be prepared to pay increasingly higher prices to obtain them.
4. **Quality and consistency of successive units of a good consumed:** We are assuming throughout that all units of a product consumed are identical. If the quality of successive chocolate bars is not the same, then the amount of satisfaction obtained from additional bars may be more or less than previous ones.
5. **Other things remain constant:** The theory assumes that all other factors affecting individuals' satisfaction remain the same. However, over time, there may be changes in income and the quality of other products as well as the development of new products and changes in individuals' tastes and attitudes, all of which may have an impact on consumers' satisfaction.
6. Having said all this, proponents of marginal utility theory argue that at any moment in time, once a particular pattern of expenditure has been established, the law of diminishing marginal utility is valid.

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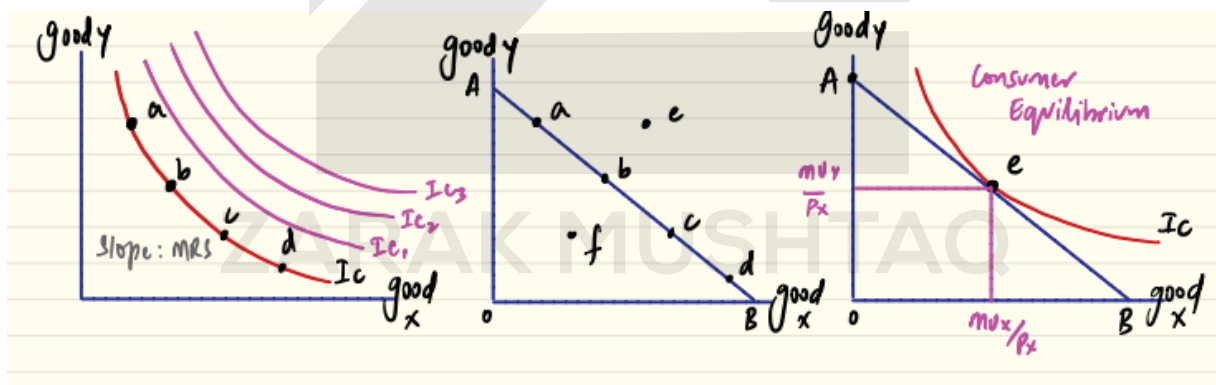
PREFERENCES AND INDIFFERENCE CURVES BUDGET LINE

ANALYSIS: The marginal utility theory approach to consumer behavior has been critical and it is not possible to measure utility. We cannot say how much the MU of one good exceeds another.

With this in mind, certain economists developed an ordinalist approach to consumer behavior. The idea was that although the consumer could not measure utility, they could order or rank their preferences. The ordering of preferences meant that the measuring of utility was not required and the theory becomes known as indifference curve analysis.

The aim of indifference analysis is to see without having to measure utility, how a rational consumer chooses between two goods. It is used to show the effects of a change in income and prices of goods and how these changes will lead to substitution and income effect.

INDIFFERENCE CURVES: An indifference curve shows combinations of products that yield the same satisfaction to the consumer i.e. it shows a set of consumption bundles among which the consumer is indifferent. The slope/gradient of indifference curve is called the **MARGINAL RATE OF SUBSTITUTION (MRS)**.



An indifference map consists of set of indifference curves. The further away the indifference curve is from the origin the higher will be the level of satisfaction it represents but along one indifference curve the level of satisfaction remains constant.

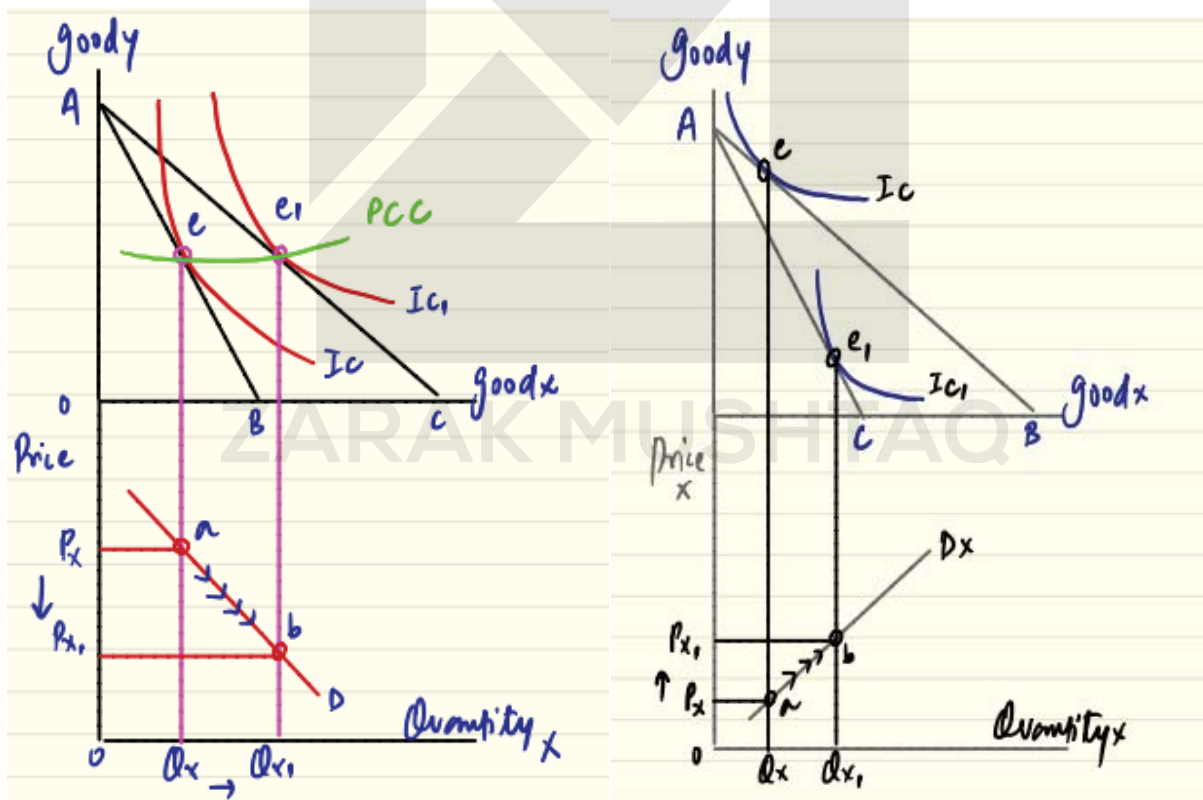
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A BUDGET LINE: A budget line shows the quantities of two goods available to consumers given his money income and the prices of the goods he buys. So, the slope of the budget line is the ratio of prices of two goods with the price of the good on the x-axis in the numerator and the price of the good on the y-axis in the denominator.

CONSUMER EQUILIBRIUM: Consumer is in equilibrium at the point where an indifference curve is tangent to budget line. At that point the slope of the indifference curve (marginal rate of substitution) is equal to the slope of the budget line (the price ratio of the goods). At this point the slope of the indifference curve is equal to the slope of the budget line i.e. $MU_x/MU_y = P_x/P_y$.

CHANGES IN CONSUMERS BUDGET: Changes in **price** will cause tilt or pivot the budget line whereas changes in **real income** can cause shift in the budget line. Real income represents purchasing power which is adjusted with inflation, thus causing parallel shift either inwards or outwards when real income changes.



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Theory of demand & supply

A **Giffen good** is a low income, non-luxury product for which demand increases as the price increases and vice versa. A **Giffen good** has an upward-sloping demand curve which is contrary to the fundamental laws of demand which are based on a downward sloping demand curve.

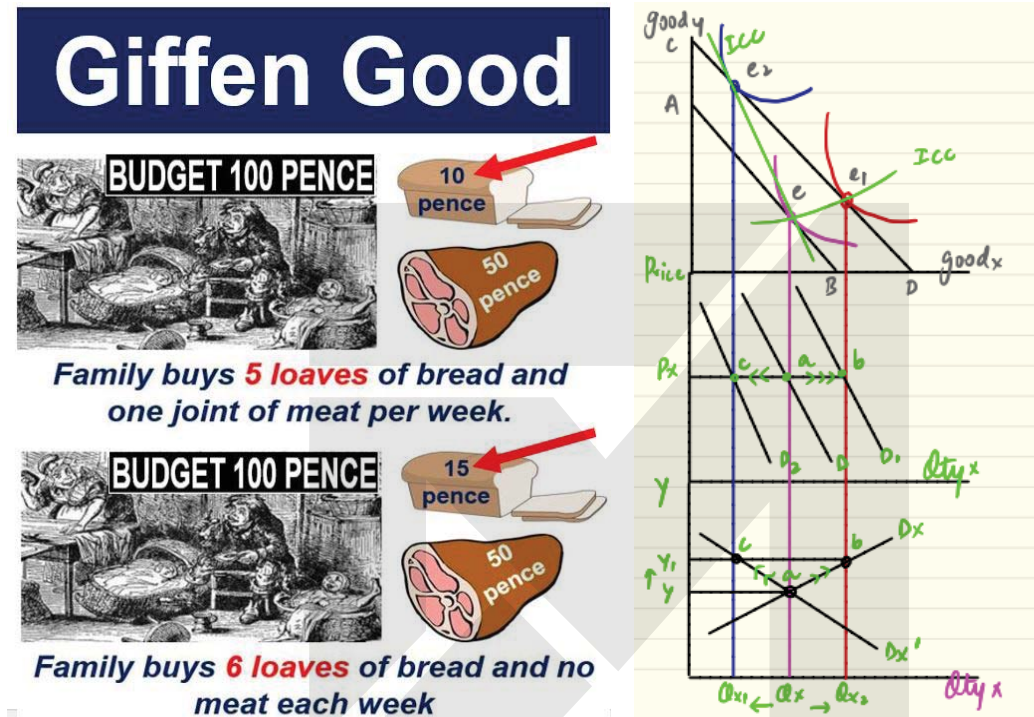


Diagram shows the effect of changes in real income, affecting the demand for normal and inferior good. An increase in income will increase the demand for normal goods while reducing the demand for inferior, that is normal goods have positive while inferior goods have negative YED. Changes in income will cause shift in the demand curve, as it is factors affecting demand other than price. Real income represents purchasing power as it is adjusted with inflation.

Income and substitution effect: Indifference curve analysis lies behind a demand curve. It can be used to examine the effect of price changes and income changes.

If the price of good X now falls the budget line will pivot. The consumer now maximizes utility consuming more of good X & good Y. The fall in the price of product X has led to an increase in quantity demanded.

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There are two different phenomena underlying a consumer's response to a price drop:

As the price of a product declines, the lower opportunity cost will induce consumers to buy more of it since it becomes less expensive - even if they have to give up other products. This is called **substitution effect**.

To isolate this effect diagrammatically we move the new budget line inwards and parallel until it is tangent to the old indifference curve. The new slope reflects the new relative prices but the utility is the same as it was originally. The substitution effect is X to X_1 . The substitution effect will always lead to more of the relatively cheaper product being demanded.

With a fixed amount of money income, a reduction in the price of a product will increase a consumer's real income - the amount of goods and services consumers are able to purchase. Typically, consumers will respond by purchasing more of the cheaper products (as well as other products). This is called **income effect**. The income effect is identified by shifting the budget line back outwards again. In this case this leads to an increase in quantity demanded of X_1 to X_2 .

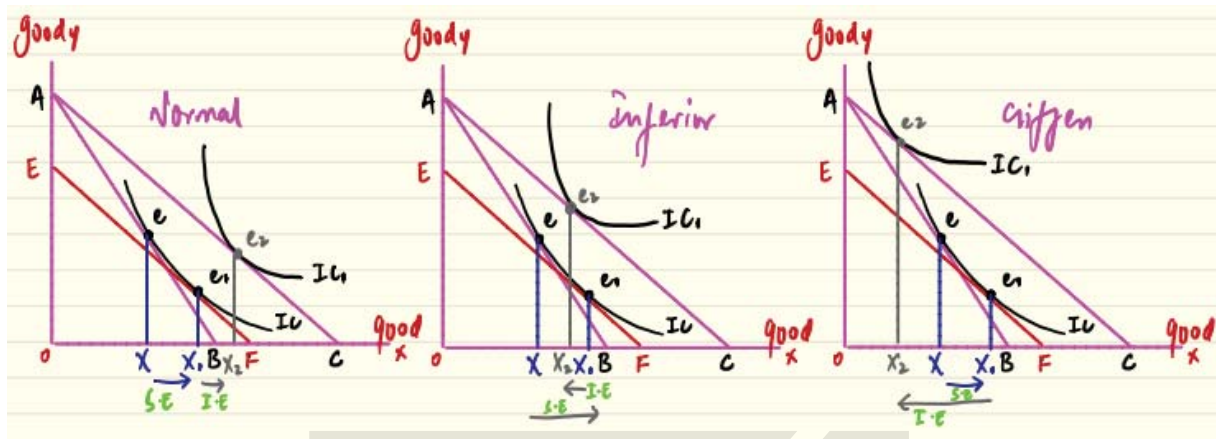
The substitution and income effects will generally work in the same direction, causing consumers to purchase more as the price falls and less as the price rises. The indifference curve can be used to separate these two effects.

In the case of a **normal good** the higher real income leads to an increase in quantity demanded; this complements the increase due to the substitution effect. This is shown in this diagram. Income and substitution effect goes in same direction in case of normal good.

In the case of an **inferior product** the income effect leads to a fall in the quantity demanded which will work against the substitution effect. In the following diagram the substitution effect is X to X_1 ; the income effect in X_1 to X_2 . The overall change in quantity demanded is an increase of X to X_2 . This means the demand curve is downward sloping because a price fall increases the quantity demanded.

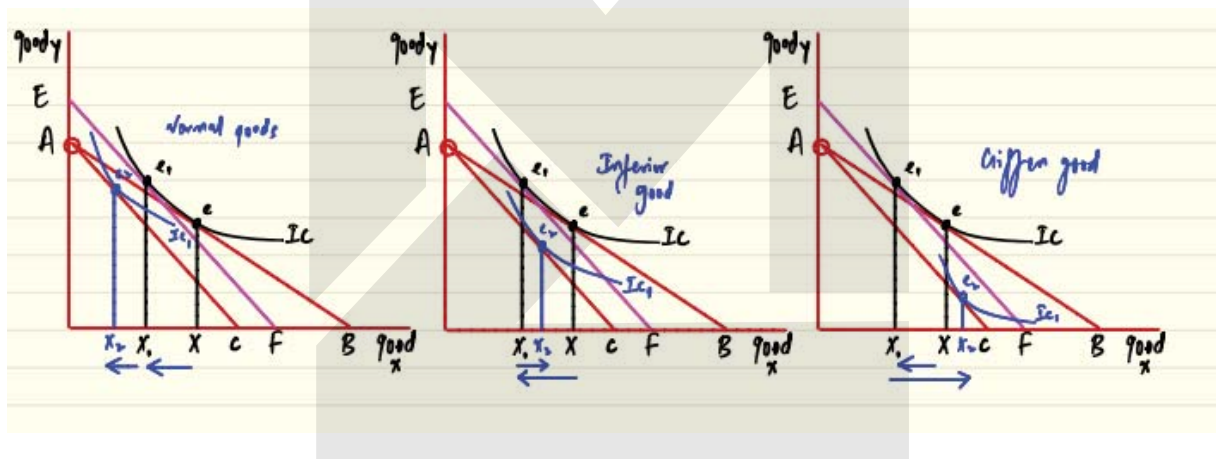
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Theory of demand & supply



In case of inferior good, income and substitution effect goes in opposite direction while in case of giffen good, income effect outweighs the substitution effect.

Income and substitution effect of a rise in price of a good:



Consumer choice and the impact of advertising: We know that people's preferences change. One obvious example is the case of advertising. If advertising is effective, it should make the advertised good more desirable. That is, it should shift consumers' preferences towards that good.

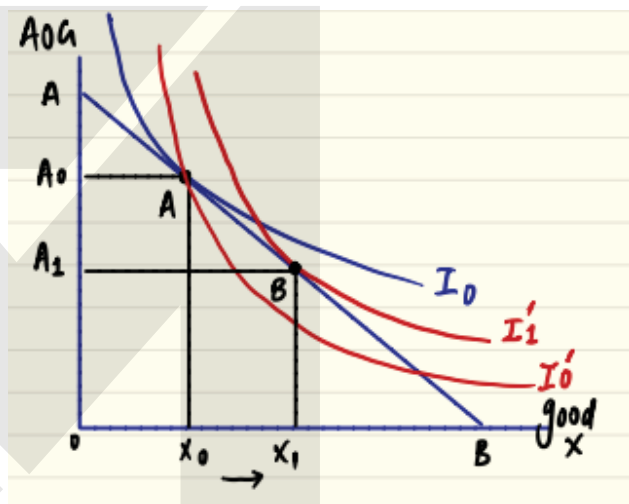
Consider the consumer choice diagram below. The consumer is initially buying the bundle of goods A, which includes XO of Good X, and AO of all other goods (AOG). Point A is on the highest indifference curve the consumer can reach (I_0) within their budget constraint (it's their best affordable choice).

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Theory of demand & supply

Now say there is a successful advertising campaign for Good X. This makes Good X more desirable. In other words, the consumer would now be willing to give up more of all other goods to buy one more unit of Good X. This means that the marginal rate of substitution (the amount of one good the consumer is willing to give up to get one more unit of the other good) has increased. The marginal rate of substitution (MRS) is the slope of the indifference curve. Since the MRS is now larger, the indifference curve becomes steeper at point A (the indifference curve pivots around to the red curve I_0').

Now, the bundle of goods A is no longer the consumer's best affordable choice. They can reach a higher indifference curve (I_1') by buying the bundle of goods B instead. The bundle of goods B includes more of Good X (X_1), and less of all other goods (A_1). So, the effective advertising campaign for Good X induces the consumer to buy more of Good X, by changing their preferences (as represented by their indifference curves).



Note that this doesn't mean that all advertising is a good idea though. Even though the consumer is now buying more of Good X (and spending more on Good X, since they are buying less of all other goods), the advertising campaign involves some cost to the firm. To determine whether the advertising campaign was a good idea, the firm would need to compare the additional revenue generated by the campaign, with its cost.

Limitations of the model of indifference curves: Although indifference curve analysis gets round the problem of measuring utility, when analyzing consumer behavior it has certain limitations.

To begin with, there is the issue of whether individuals can realistically be expected to be able to identify imaginary combinations of two goods which will give equal amounts of satisfaction and then compare them with other combinations of the same two goods which give more or less satisfaction, which is what is necessary

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Theory of demand & supply

to create an indifference map. It is also the case that indifference curves are drawn on the basis of an individual's perception of the amount of utility that will be obtained from commodities, which might differ significantly from the satisfaction that is obtained when they are actually consumed. Moreover, indifference curve analysis is unable to explain purchases of goods such as cars or computers or other consumer durables which are purchased frequently with no consistent trade off against another product. Such goods may also only be purchased once. Finally, consumers may not act rationally when planning their expenditure. This latter point leads us into a developing branch of economic analysis known as behavioral economics.

Behavioral economics: A major assumption underpinning both marginal utility theory and indifference curve analysis is that consumers always act rationally. In the sense that economists use the term, rational behavior in mainstream Economics means that:

1. individuals act solely to maximize their own personal utility (satisfaction)
2. individuals are in possession of or have access to all the information they need at low or zero cost
3. consumer decisions are based upon a careful comparison of the costs and benefits in order to achieve the optimum final outcome
4. once behavior has been optimized, decision-making is based on changes at the margin
5. Individuals' preferences and attitude to risk are fixed.

In the real world, however, it is clear that individual behavior may be irrational from the point of view of traditional economic models. For example, in financial crises and stock market crashes over the past century, investors and financial institutions appear to have repeated the mistakes of the past. At the individual level, why do individuals keep substantial surplus funds in current accounts or underperforming savings accounts when other accounts paying higher rates of interest are readily available? Why is it that many individuals automatically renew their annual car insurance with their existing company when identical cover can be obtained elsewhere at a lower price?

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Behavioral economics is a rapidly developing branch of Economics which attempts to explain such apparently "irrational" behavior. Relying on empirical evidence and experiments, behavioral economics incorporates findings and insights from other disciplines such as sociology, social anthropology and particularly psychology to try to explain how individuals actually behave.

The pioneering study in behavioral economics is "Prospect Theory: An Analysis of Decision under Risk" (1979) by Daniel Kahneman and Amos Tversky.

Prospect theory attempts to explain how people choose between alternatives involving risk where the probability of each outcome is known. It argues that rather than attempting to achieve optimal outcomes as traditional economic theory suggests, individuals actually make decisions based on the potential gains and losses that may arise.

Kahneman, Tversky and other behavioral economists identify a number of key elements in explaining individuals' apparently irrational behavior.

Bounded rationality and heuristics: Let us return to our earlier example of irrational behavior - all individual automatically renewing their annual car insurance with the existing provider. There is obviously all enormous quantity of information available which would enable the individual to make a rational decision as to whether this is in fact the best decision. There are many insurance companies from which to buy, each with their own publicity, website and telephone Centre's, as well as many comparison sites. However, for a variety of reasons individuals may not be able to use this information effectively to arrive at an optimum outcome. To begin with, the timescale may be short and accessing the quantity of available information may take many hours and possibly even involve some financial cost. The individual may feel that the opportunity cost in terms of the time taken in sifting through the information and inputting the same personal information possibly many times may not be worth any potential savings. They might simply find the enormous volume of information overwhelming - too much to absorb within the available timescale. The individual's ability to act in a rational manner is, therefore, restricted; in the language of behavioral economics it is "bounded". In the case of making other decisions, it could be the lack of sufficient information or the fact

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Theory of demand & supply

that there may be more than one possible solution that means a consumer may experience what Herbert Simon (1978) termed bounded rationality. As a result, individuals may engage in satisficing rather than optimizing behavior in their decision-making.

Heuristics: In such cases, behavioral economists argue that individuals resort to heuristics, which are simplifying strategies or mental shortcuts to arriving at solutions. They argue that empirical evidence and experiment identify many such heuristics including the following:

1. **Anchoring:** the tendency to rely on the first piece of information obtained (the anchor) when considering a decision.
2. **Availability:** basing decisions on the easiest piece of information to recall. The way and order in which information is presented to an individual will be important here.
3. **Representativeness:** basing decisions on past experience or assumptions. This can lead to stereotyping. An example of the representativeness heuristic would be to argue that simply because we have only met low-income individuals from a particular region that all people from that region must be poor. This particular example illustrates that although the use of these heuristics (rules of thumb) may speed up decisions, it can also lead to individuals making mistakes or irrational decisions.

Framing: Another feature of individual decision-making appears to be what behavioral economists refer to as framing. The way in which an issue, question or choice is presented or framed. There is evidence that the same information presented in different ways may lead to a different decision being made. For example, an individual may be more likely to buy an expensive pair of designer shoes if they are advertised as reduced from \$400 to \$300 rather than simply being priced at \$300. Retailers can obviously use this knowledge in advertising and marketing their products.

Implications for policy Behavioral economists argue that their findings can have important implications for government policy. Richard Thaler and Cass Sunstein in "Nudge: Improving Decisions About Health, Wealth and Welfare"

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(2008) suggest that insights from behavioral economics can be used by governments and others to persuade or "nudge" people into making decisions which are beneficial for them or in their best interests. For example, they suggest that individuals will be more likely to adopt energy conservation methods in their homes if the government advertises them as leading to a saving of \$X per year rather than stating that they will lose (the same) \$X per year if they do not adopt them - an example of "framing"

THE THEORY OF SUPPLY

PRODUCTION: Any activity which is designed to fulfill the wants of people is called production. Production process completes when the commodity is in the hands of the final consumer. Hundreds of inputs enter the production of any specific output. Among the many inputs entering into car production, for example, the steel sheet, rubber, spark plugs, electricity, the site of the factory, mechanics, cost accountants, spray-painting machines, forklift trucks, painters, managers etc. In order to produce the goods and services that it sells, each firm need inputs. These can be grouped in to four broad classes:

1. Those that are man-made aids to production, such as spark plugs, electricity, steel sheet, factory building, machines.
2. Those that are directly provided by the nature, such as land.
3. Those that are directly provided by the people, such as services of managers and workers.
4. Those who organize the rest of the factors of production, such as the owners of the business.

The outcome of the production process is called output which is divided in to:

- Consumer goods which directly satisfy the wants of the people. These are further divided in to consumer durable (which last for a long time, usually more than one year, e.g. cars, furniture etc.) and non-durable goods (which last for less than a year, e.g. food, soap etc.).

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Theory of demand & supply

- Producer goods which do not directly satisfy consumers want. They are bought by the producers to produce goods and services e.g. machines and raw materials.
- Services are intangible commodities e.g. banking, insurance, transport.

THE SHORT RUN PRODUCTION FUNCTION: The short run is defined in economics as a period of time where at least one factor of production is assumed to be in fixed supply i.e. it cannot be changed. We normally assume that the quantity of capital inputs is fixed and that production can be altered by suppliers through changing the demand for variable inputs such as labor etc. often the amount of land available for production is also fixed.

The short run period of time differ from industry to industry. The short run for the electricity generation industry or the telecommunications sector varies from that appropriate for newspaper and magazine publishing and small scale production of foodstuffs and beverages. Much depends on the time scale that permits a business to alter all of the inputs that it can bring to production.

A firm **production function** shows that maximum output it can produce with any given set of inputs. It relates inputs to outputs and describes the technical relation between what is fed into the production process by way of materials and the inputs of factor services and what is turned out by way of product.

PRODUCTION FUNCTION: $Q = f(K, L)$

Where Q is the quantity of output of some good or service and K and L are the quantities of capital and labor.

SHORT RUN CHANGES IN OUTPUT AND THE LAW OF VARIABLE

PROPORTIONS OR LAW OF DIMINSHING RETURNS: To increase the output in the short run, when the producer employs the variable factors of production with the fixed factors of production, the output increases in a different proportion as compared to the increase in the factor of production. This led to the formulation of the famous law of diminishing returns.

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Assumptions: A particular crop is grown on a fixed area of land of 20 acres. The amount of capital is fixed in supply and the labor is the variable factor.

Assumptions on which the example is based are:

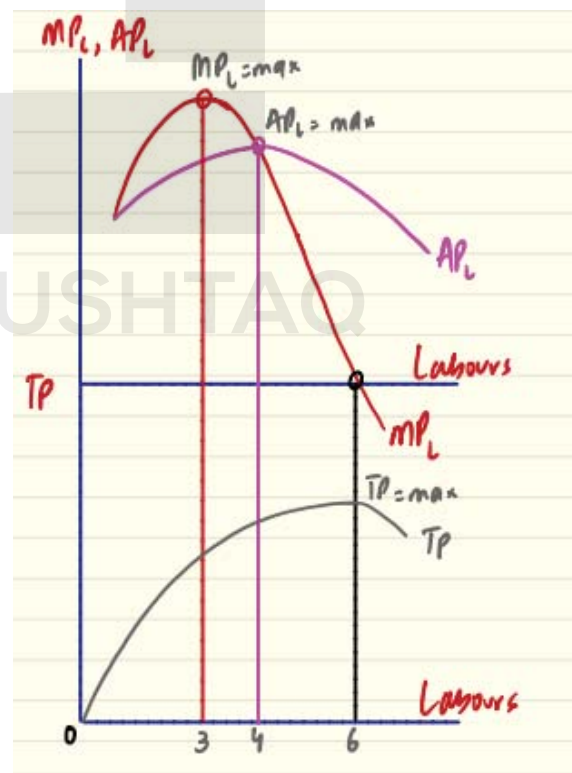
- Labor is the variable factor of production
- All the labors are homogenous i.e. they have similar mental and physical capacities
- Technology remains constant

On these assumptions it can be concluded that any changes in productivity arising from variations in the number of people employed are entirely due to the changes in the proportion in which labor is combined with the other fixed factors.

The law of diminishing returns states that as we add more units of a variable input (i.e. labor or raw materials) to the fixed amount of land and capital, the change in total output will at first rise and then fall. Diminishing returns to labor occurs when the marginal product of labor starts to fall. This means that the total output will still be rising - but increasing at a decreasing rate as more workers are employed.

What happens to marginal product is linked directly to the productivity of each extra worker employed. At low levels of labor input, the fixed factors of production - land and capital, tend to be under-utilized which means that each additional worker will have plenty of capital to use and, as a result, marginal product may rise. Beyond a certain point however, the fixed factors of production become scarcer and new workers will not have much capital to work with so that the capital input becomes diluted among a larger work force.

TOTAL PRODUCT: It is the total amount



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Theory of demand & supply

of goods and services produced during some period of time by all the factors of production that the firm uses. As more labors are employed the output increases.

AVERAGE PRODUCT: It is the total product per unit of labor or labor productivity

$$AP = TP/L$$

MARGINAL PRODUCT: It is the change in total product resulting from the use of one more unit of the variable factor. MP at first rises reaches some maximum and then declines. The level of output where MP reaches a maximum is called the point of diminishing marginal returns.

$$MP = \text{change in TP} / \text{change in L}$$

TP = total product

L = labor

The shape of TP curve shows that initially it increases at an increasing rate and then it increases at decreasing rate.

Relationship between AP and MP: Since the labor is the only variable factor in the example, changes in output are related directly to changes in employment. So it is possible to speak of a changes in the productivity of labor or changes in the return to labor. As the number of people increases, total output continues to increase, but this is not true of the average and marginal products. As more labor is employed both AP and MP begins to rise, reach a maximum and start to fall. The AP curve slopes upwards (rising) as long as the MP curve is above it; no matter MP curve it-self is rising or falling. The common sense of this relationship is that if an additional labor is to raise the AP of all the labors, the labor's output (MP) must be greater than the average output of all the labors. Since, AP rises or falls depends on whether MP is above or below AP, it follows that MP must equal AP at the highest point on the AP curve.

LONG RUN THEORY OF PRODUCTION: In the long run the producer increases all the factors of production to operation a larger scale of production. When an increase in the scale of production results in more than proportionate

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Theory of demand & supply

increase in output, the firm is said to be experiencing economies of scale (increasing returns to scale) which lead to falling long run average costs.

Internal economies of scale arise from an increase in the scale of production in the firm itself by increasing the number of plants or increasing the size of the existing plant. External economies of scale are those advantages which a firm gains from the growth of industry or because of concentration of firm in one geographical area.

INTERNAL ECONOMIES OF SCALE:

1. TECHNICAL ECONOMIES

- (a) Increased specialization: Large scale producers can effectively divide the complex production process into many separate operations and each operation is performed by a particular group of workers. This will increase their productivity thereby causing per unit to fall.
- (b) Indivisibility: Some capital equipment, such as blast furnace, are so large that they can be operated efficiently only when there is bulk production, causing per unit to fall.
- (c) Increased dimensions: Firms which make use of tanks, transport equipment, warehouses etc. can increase their volumes many times the increase in their dimensions. So the proportionate increase in the volume is much more than the increase in the costs causing per unit cost of storage or transporting goods reduce significantly.
- (d) The principles of multiples: A production process includes different machines with different processing speeds. Small scale producer can have one machine of each type and will operate the slower machines inefficiently. But the large scale producer can have a combination of all machines such that all the machines process equal number of units per period of time, thereby making efficient use of machines.
- (e) Economies of linked processes: A large plant may have the capacity to produce more than one product e.g. steel is produced by reprocessing iron in the same blast furnace.

- 2. PURCHASING ECONOMIES: Large scale producers buy raw material in bulk which enable them to receive attractive discounts which significantly reduce

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Theory of demand & supply

per unit cost. Large firms are able to employ specialized buyers who have better skills and knowledge which enable them to buy the right materials at the right price. Because of bulk production per unit selling costs of large firms are much lower than the small firms, similarly per unit costs will also be lower for the large scale producer.

3. **FINANCIAL ECONOMIES**: Large scale producers have many external sources of finance than small scale producers. The fact that the firm is large and well known makes it more credit worthy borrower. Its great selling potential and larger assets provides the lenders with greater security and encourages them to provide bigger loans at lower interest rates than it would charge to a smaller firm.
4. **RESEARCH AND DEVELOPMENT ECONOMIES**: Large scale producers usually have significant finance to have a research and development department of their own which help them to make improvements in their products and to reduce the per unit costs.
5. **RISK BEARING ECONOMIES**: Large scale producers sell goods throughout the economy where increases and decreases in demand for the good cancel out each other thereby the total demand, production and price of the product remain more or less stable. Many large firms choose to reduce the risks of trading by means of a policy of diversification. They produce either a variety of models of a particular product or a variety of products so that they do not put all their eggs in one basket. A fall in demand for any one of its products may not mean serious trouble for the firm.
6. To diversify the product range. A multi-product firm has the advantage of being able to spread business risks. If one branch of its activity is stagnating or going into decline, there will still be the revenue from others to keep the firm afloat. Firms often see new business opportunities in related areas. This is at the heart of successful entrepreneurship. Sometimes large firms can use the same production facilities, so keeping the costs down. These benefits are called **ECONOMIES OF SCOPE**

EXTERANAL ECONOMIES OF SCALE: It arises from growth in the size of the industry i.e. they are external to the firm. As an industry grows in size, certain benefits become available to all the firms in the industry. These benefits are

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Theory of demand & supply

particularly important where the industry is heavily localized, that is, concentrated in a particular region.

1. **A SUPPLY OF SKILLED LABOR:** Where an industry localized, the supply of labor in that area becomes skilled in the trades and adapted to the working conditions of that industry. If a firm joining the industry were to set up its factory away from the established Centre of the industry, it would have to meet the costs of training people in the skills required.
2. **DISINTEGRATION:** When an industry reaches a certain size, firms often tend to specialize in a single process or in the manufacture of a single component. When this happens the industry is said to have disintegrated. For example, car manufacturers assemble cars; they do not make every part of a complete car. Many of the components are purchased from specialist producers. Firms making wheels, doors, brake linings etc. for the entire industry can produce these components on a much larger scale than any single car manufacturer. All the firms in the industry, therefore, are able to obtain components at lower cost than they could produce these articles for themselves.
3. **SUBSIDIARY INDUSTRIES:** When a large industry is concentrated in a particular region, subsidiary industries establish themselves in the locality to cater for the needs of all the firms in that industry. Thus in the area around Northampton, where there are many footwear manufacturers, we find firms which specialize in the production of shoe making machinery and in the maintenance and repair of this equipment.
Other examples of subsidiary industries are those which specialize in making use of waste materials of a large industry and those which produce containers and packaging materials specially designed to meet the requirements of firms in a major industry.
4. **SPECIALIZED SERVICES:** When a heavily localized industry reaches a certain size, it becomes worthwhile for firms to supply specialized services to that industry. Such firms may provide wholesaling services, transport services, research and information services, and financial services and so on. These services will be designed to meet the particular needs of firms in a

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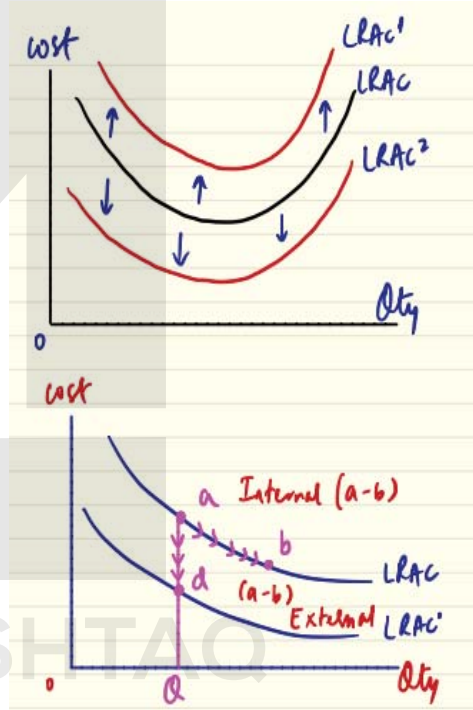
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large localized industry. It would be very costly for the individual firms in the industry to supply all these services for themselves.

Local banks and insurance companies will become familiar with the need of the firms in that industry, and offer services which are suited to their requirements.

DISECONOMIES OF SCALE (decreasing returns to scale): The list of advantages of large scale production is very impressive, but growth in the size of the firm can also create problems. For any particular industry, there seems to be an optimum size of firms, where average cost is at a minimum. When firms grow beyond the optimum size, diseconomies of scale are experienced that cause long run average cost to rise.

INTERNAL DISECONOMIES OF SCALE: The main cause of these diseconomies appears to be increasing problems faced by management as firm grows larger. Large firms are usually divided in to several specialist departments. There are departments responsible for sales, production, stock control, accounting, buying, personal and so on. It is not an easy task to organize and supervise the activities of several large departments and to make sure that these departments are working together as a team. Weakness in organization and control can be a cause of inefficiency and hence increasing costs.



A major challenge facing the managers of large firms is to find ways of obtaining the willing cooperating of the workers. Individual workers in a labor force of several thousand may find it difficult to identify themselves with the firm; they do not feel that they are part of the firm. Where management is weak and inefficient, workers may come to believe that as far as the firm is concerned, they are little more than the numbers on the wage sheet. Such a state of affairs is not likely to lead to efficient and low cost production.

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EXTERNAL DISECONOMIES OF SCALE: A firm may also experience disadvantages as a result of the industry to which it belongs becoming too large. These are referred to as diseconomies of scale. For example, a shortage of labor with the appropriate skills may develop so that firms in the industry may find themselves bidding up wages as they try to attract more labor. Increasing demands for raw materials may also bid up prices and cause costs to rise. If the industry is heavily localized, land for expansion will become increasingly scarce and hence more expensive both to purchase and to rent. Transport costs may also rise because of increased congestion. All firms in the industry, whether they are seeking to expand or not, may suffer rising costs as a result of the industry getting too large too quickly.

WHY FIRMS GROW?

- To **reduce average cost**: The most obvious and sensible reason why firms might try to grow is to obtain economies of scale and reduce their costs of production.
- To obtain a **larger share of the market**: The larger the firm, the larger the market share. The desire to exercise a greater control over the total markets is an important reason for growth in the size of a firm.
- To achieve **greater security**: Larger firms, with their greater resources and wider range of products, are better equipped to withstand the fluctuations in demand. This is an important reason for the development of firms producing different products. The total demand for a variety of products is likely to be more stable than the demand for any one of these products.

HOW FIRMS GROW?

There are two methods by which growth may be achieved. The first of these is the **internal growth**, where a firm increases its size by making more of its existing product or by extending the range of its products. It grows within the framework of its existing management and control structure.

The second and more common method nowadays is **external growth**. This is achieved by merger or takeover. A merger usually implies consent, with two or more firms agreeing to combine to create a few firms whereas a takeover is

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sometimes taken to mean that the firm over loses its identity and is absorbed into the firm carrying out the takeover.

- **Vertical integration:** This term describes the joining together of firms at different stages in the process of production. It may take form of backward integration or forward integration. **Vertical integration backwards** takes place where a firm takes over or merges with its suppliers or the previous stage of production. Thus, a firm making a chocolate may take over a cocoa plantation, or a car manufacturer may acquire a body-building plant. A firm is likely to integrate backward because it wishes to have a greater degree of control over the quality of its supplies and regularity of its delivery. **Vertical integration forwards** occurs when a firm takes over its market outlets or the next stage of production. Thus, a manufacturing firm might acquire a chain of retail shops which sell its products, or an oil company may take over a number of petrol stations. Forward integration is often carried out when producers wish to improve the quality of the premises. The major oil companies have obtained control over many petrol stations and have invested heavily on schemes to modernize their outlets.
- **Horizontal integration:** This form of integration takes place when firms producing similar products are joined together to form one organization. The amalgamations of two or more mining companies, or the merger of firms owning chains of clothing shops, would be described as horizontal integration.

There are several reasons for this type of integration but the most important is to achieve greater market share to exercise monopoly in the market. This is done by merging with or taking over the competitors thereby reducing the competition. The other important reason of horizontal integration is to achieve the economies of scale and to reduce per unit cost. Horizontal integration may also be done to **rationalize** the industrial capacity. This term is used to describe the process of eliminating less efficient plants or factories, and concentrating production in the more efficient plant. For example there are there firms producing the similar product. Each firm is producing one third of its capacity. If one of the firms takes over the other two firms and produces the product on one plant

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at full capacity and shut down the other two then the total output will be the same but the per unit cost will reduce significantly.

- **Conglomerates:** Sometimes amalgamations take place between firms which do not produce similar products and which are not part of the same production process. This type of integration is described as a conglomerate merger. The firms which join together belong to very different industries, and use different raw materials and different technical processes.

An important reason for this type of integration appears to be the desire for **greater security**. A conglomerate will own companies producing very different goods; hopefully, the success of one company will tend to offset the lack of success of another company.

Although two firms may be producing very different products, there may be links between them which provide a good reason for integration. For example a soft drink producer may emerge with a potato chips producer because products are distributed through the same kind of retail outlets.

Multinational firms: A multinational firm is a company which produces products in more than one country. Multinational companies are sometimes referred to as multinational corporations or transnational corporations. Most MNCs come from industrialized countries, particularly the USA. They are very significant with the combined output of the largest two hundred MNCs being equivalent to approximately one third of the entire output of the world. Indeed the output of General Motors exceeds the output of many countries including Denmark, Norway and Poland.

Advantages of operating abroad: Producing and selling in a major market abroad will reduce a firm's transport costs, enable it to keep in close contact with that market and may enable it to overcome any resistance to buying products made outside the host country. Firms also set up plants abroad in order to avoid import restrictions, take advantage of regional assistance or to obtain new materials. In some cases MNCs set up plants where wages are lower

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Theory of demand & supply

The survival of small firms

Market limitation on mass production: The ability to take advantage of large scale production is limited by the size of the market. Where the market is small it is not possible to obtain significant economies of scale, most of which are dependent upon the existence of a large market for a standardized product. The size of the market may be restricted by a number of factors.

- **A demand for variety:** Some industries are faced with the problem that consumers demand a wide variety of styles, patterns and designs. The market for any one style or model may be very small indeed. Obvious example is the clothing, millinery, furniture, footwear and jeweler trades. Firms in industries catering for such variety are confronted with the problem of 'short runs' and 'one-off' types of production - they cannot set up their capital and labor for specialized production. In such cases firms are likely to be small.
- **Geographical limitation:** If a product has great bulk in relation to value, transport costs will be high relative to production costs. In such cases the market for the product is likely to be local rather than national. Bread, bricks and coal may be cited as examples of products where markets have been confined to fairly small regions, although improvements in transport have tended to weaken this particular restriction on the size of the firm. Transport costs, however, still provide the village or suburban shop with some protection against out-of-town superstore.
- **Personal services:** Industries which provide services rather than manufactured goods are usually characterized by a large number of small firms. Where the element of personal attention is required by the purchase is an important part of the service, and then it is not possible to introduce standardization and mass production methods. As a result a large number of small firms are found in profession such as law, accountancy, architecture, and medicine where personal attention to individual problems is required. For similar reasons most firm's specializing in repair work are relatively small. Hairdressing and tailoring provide personal attention seriously limiting the size of the firm.

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Theory of demand & supply

- **Luxury items:** The market may be limited by income and wealth. Expensive sports cars, luxury yachts, high quality jeweler and fur coats are examples of goods produced by small firms for very restricted prestige markets.
- **Disintegration:** As already noted there is a tendency for mass production industries to disintegrate into a large number of specialist firms each supplying some standardized part to a large assembly plant. This has been structural change which has operated very much to the benefit of the firm. It is possible for the industry's total requirements of some particular component to be supplied by one relatively small firm.

THE LONG RUN EQUILIBRIUM OF THE FIRM: In the long run all the factors can be varied. The firm can thus choose what techniques of production to use; what design of factory to build, what type of machines to buy, how to organize the factory, whether to use highly automated processes or more labor intensive techniques.

The profit maximizing firm will obviously want to use the least cost combination of factors to produce and given level of output. It will therefore substitute factors, one for another, if by doing so it can reduce the cost of a given output. If further substitution between the factors does not reduce per unit cost, the producer is producing the output at the minimum possible cost with the given state of technology.

Such profit increasing substitutions are always possible whenever the marginal product of the one factor per \$ spent on it is greater than the marginal product of the other factor per \$ spent on it. The firm has not minimized its costs as long as these two magnitudes are unequal. For example, if an extra \$ spent on labor produces more output than an extra \$ spent on capital, the firm can reduce costs by spending less on capital and more on labor i.e. $MP(L) / MP(L) > MP(K) / MP(K)$ more labor should be used relative to capital, since the firm is getting a greater physical return for its money from extra workers than from extra capital. As more labor is used, diminishing return to labor set in. thus, MP of labor will fall. Likewise, as less capital is used per unit of labor, MPk will rise. This continue until:

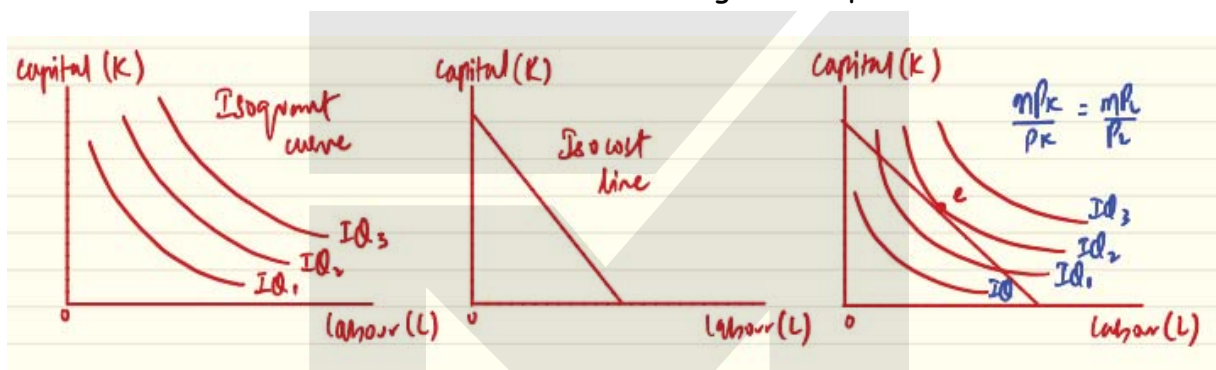
$$MP(L) / MP(L) = MP(K) / MP(K)$$

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Theory of demand & supply

At this point the firm will stop substituting labor for capital or vice versa. Thus, the firm achieves the long run equilibrium where marginal product per pound spent on each factor is the same.

ISO-COST, ISO-QUANT APPROACH: An isoquant shows different combinations of two factors (e.g. capital and labor) which produce the same level of output i.e. output remains same throughout an isoquant. A higher isoquant shows a high level of output. The slope of the isoquant is called MARGINAL RATE OF FACTOR SUBSTITUTION which reduces as we move along the isoquant curve.



$MRS = \text{change in } K / \text{change in } L$

Along an isoquant the change in output will be zero (rise in Q = fall in Q).

An isocost line shows the combinations of two factors which have the same cost of production. The slope of an isocost line shows the change in factor combination. It means that the fall in the expenditure on capital is equal to the rise in expenditure on labor. The producer is in equilibrium at a point where the isoquant is tangent to isocost line.

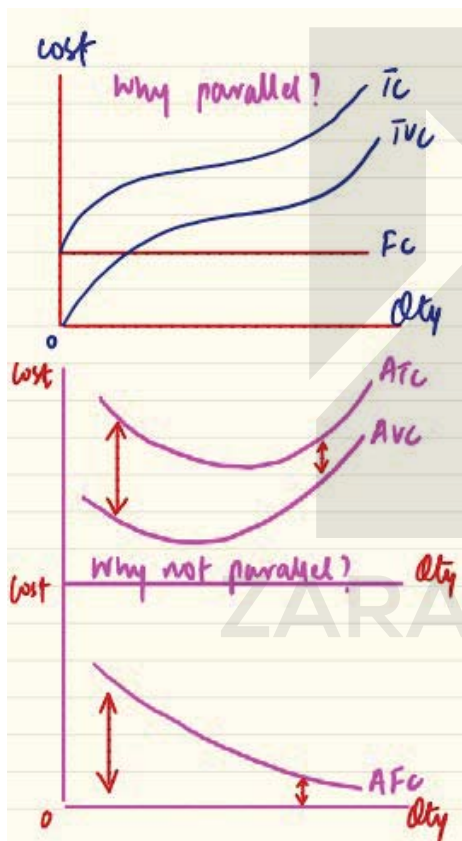
COSTS OF PRODUCTION: In dealing with the relationship between output and costs there are two circumstances to consider. First, short-run changes which cover periods when it is only possible to adjust the amount of the variable factors of production and where at least one factor is fixed. Secondly, long-run changes which apply to periods of time which are sufficiently long for all the factors of production to be changed.

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Theory of demand & supply

Costs in the short run: When measuring costs, economists always make use of the opportunity costs i.e., what the producer has to forgo to produce output. These costs are divided as implicit and explicit.

Explicit costs are those opportunity costs of factors not readily owned by the firm. These are the prices which the firm pays for the fixed and variable factors employed by the firm. Implicit costs are the hidden cost of production i.e. the opportunity costs of the factors which the firm already owns. They are equal to what the factors could earn in some alternative uses e.g. the return on capital employed in alternative uses, the interest forgone on the money which the producer uses into his business for his past savings. When we speak of the total costs of the firm we mean the sum of implicit and explicit cost.



FIXED COSTS: These are the costs associated with the fixed factors of production and do not vary with the output. For example, rent, rates, insurance etc. whether the firm operates at full capacity or less than full capacity, these cost have to be paid.

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Theory of demand & supply

Let us look at a business such as the Suzuki Motor Company. The company has big machines, thousands of parts, huge buildings and a multitude of other pieces of plant and equipment that have already been bought. Suzuki has to take account of the wear and tear of this equipment no matter how many cars the company produces. The payments on the loans taken out to buy the equipment and the rent of the land have to be paid regardless of output. All these costs are unaffected by variations in the quantity of output, i.e. they are mainly the overhead costs or fixed costs.

VARIABLE COSTS: These are the costs associated with the variable factors of production i.e. they vary with the production. They are sometimes referred to as operating costs e.g. wages of labor, costs of raw materials, transport cost and other utilities. When output is zero there is no variable cost incurred but as the producer increases output in the short run the variable costs start increasing with the output.

The **shape of the VC** follows the law of diminishing returns. Initially variable cost rises sharply because the fixed factors are heavily underutilized and then VC rises less and less rapidly as more and more variable factors are employed. The reason being that the division of labor and specialization make a fuller utilization of capital equipment and increasing return of the variable factors are achieved. After certain level of output VC start increasing rapidly because of diminishing return.

TOTAL COST: It is simply the sum of the fixed and the variable costs. The shape of the total cost curve is the same as the variable cost curve with a vertical distance equal to the fixed cost.

SHORT RUN AVERAGE COST CURVES:

The average cost concept is simply the one in which we are measuring the cost per unit of output.

- Average total cost = total cost / output
- Average variable cost = total variable cost / output
- Average fixed cost = fixed cost / output

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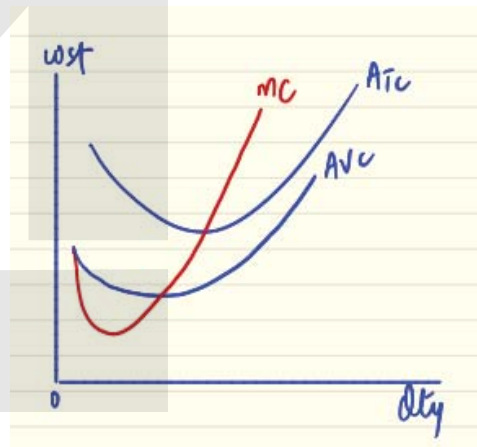
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AVERAGE FIXED COST: It is the fixed cost per unit of output. AFC continue to fall throughout the output range

AVERAGE VARIABLE COST: AVC measures the variable cost per unit of output. AVC declines initially, reaches some minimum and then increases again. Graphically it is a U-shaped curve. In simple terms, at low levels of output, production is relatively inefficient and costly, because the firm's fixed plant is understaffed. Not enough variable resources are being combined with the firm's plant; production is inefficient and per unit variable costs are therefore relatively high. As output expands, however, greater specialization and better utilization of firm's capital equipment will yield more efficiency and variable cost per unit will decline. As more variable resources are added, a point will be reached where diminishing returns are incurred. The firm's capital will now be staffed more intensively and therefore each added input will not increase output by as much preceding inputs. This means that AVC will eventually increase.

AVERAGE TOTAL COSTS: It measures the cost per unit of output and is the sum of AFC and AVC.

MARGINAL COST: It is defined as the change in total cost brought about by one unit change in output, i.e. the cost of an extra unit of output produced.



Marginal costs = $\text{change in TC} / \text{change in Q}$

The only way for AVC to rise is that the MC should be greater than the AVC. There is also a similar relationship between ATC and MC i.e. when MC is greater than ATC, it will cause ATC to rise and when MC is less than ATC, and it will cause ATC to fall. Marginal cost will intersect the ATC and AVC at their minimum point. The minimum of AVC lies at relatively lower level of output than the minimum of ATC. This is because ATC also includes AFC which keeps on falling with the increase in output.

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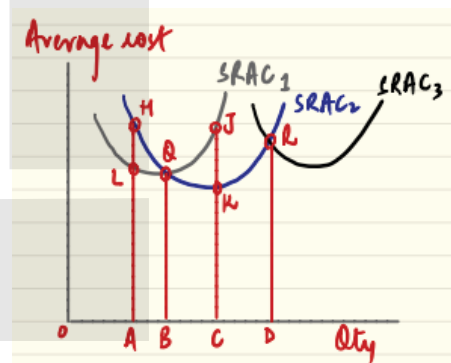
THE LONG RUN AVERAGE COSTS: In the long run there is no distinction between fixed and variable costs because all the factors are variable. So there is a long run average cost for any given set of input prices.

1. Factor prices are constant: Along a LRAC curve the prices of the factors of production are constant and for any given level of output a change in factor prices will shift the LRAC
2. The state of technology and the factor quality are constant: Technology changes in the very long-run and when a firm gains EOS, it is actually exploiting the existing technology in a better way.
3. Firms choose the least cost combination of the factor of production: Along a LRAC the firm operates efficiently i.e. it will satisfy the cost minimizing condition.

If the firm does not choose the cost combination of the factors of production then it will be producing at a point above LRAC curve. There are many short run curves as a firm develops over the years but only one LRAC curve. LRAC therefore, is also called as a planning curve.

Deriving a Long Run Average Cost Curve:

To understand the derivation of a long run average cost curve, let's consider three short run average cost curves (SACs) as shown in Fig. 1 below.



These SACs are also called plant curves. In the short run, a firm can operate on any SAC, given the size of the plant. For the sake of our understanding, let's assume that there are only three plants that are technically possible. Therefore, the firm increases or decreases its outputs by changing the amount of the variable inputs. However, in the long run, the firm examines each SAC to find the curve that allows it to produce a given level of output at the minimum cost. Hence, it chooses between SAC_1 , SAC_2 , and SAC_3 . From Fig. 1 above, you can see that to generate OB amount of output, the firm can choose between SAC_1 and SAC_2 . Note that the firm will choose SAC_1 due to the lower costs as compared to SAC_2 .

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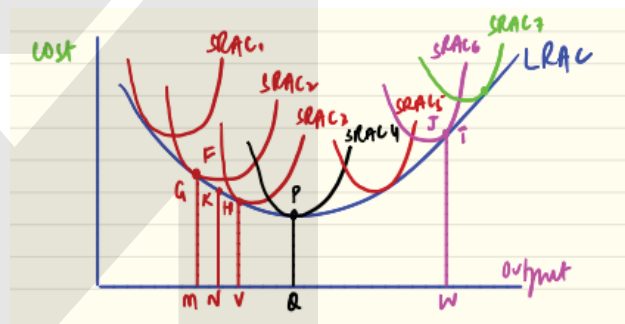
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Further, you can also see that if the firm tries to produce an output OA , then it costs

- AL per unit with SAC_1
- AH per unit with SAC_2

Clearly, $AH > AL$. Therefore, the firm chooses SAC_1 . Similarly, if the firm tries to produce an output which is greater than OB but less than OD , then it chooses SAC_2 since SAC_1 involves higher costs. Also, for outputs larger than OD , the firm uses SAC_3 . Summing up, we can say that in the long run, the firm employs the plant yielding maximum output at minimum cost per unit.

Long Run Average Cost Curve: Imagine if a firm has a choice of varying a plant by infinitely small gradations leading to infinite average cost curves. In such a case, the smooth curve enveloping all these short-run average cost curves is a long run average cost curve.



As you can see in the figure above, the long run average cost curve is drawn tangential to all SACs. In other words, every point on the long run average cost curve is a tangent point on some SAC. Hence, whenever a firm desires to produce a certain output, it operates on the corresponding SAC. From the Fig. 2 above, you can observe that to produce an output OM , the corresponding point on the long run average cost curve is 'G'. Also, the corresponding SAC is SAC_2 .

Therefore, the firm operates on SAC_2 at point G. Similarly, the firm chooses different SACs based on its output requirement. It is also possible for the firm to produce the output OM with SAC_3 . However, this will lead to a higher cost of production as compared to SAC_2 . On the other hand, to produce a higher output OV , the firm requires SAC_3 . If the firm uses SAC_2 for the same, then it results in higher unit similarity.

TOTAL REVENUE: It is the total earnings of a firm from the sale of given quantity of goods and services; it is calculated by multiplying the price of the good with the quantity sold.

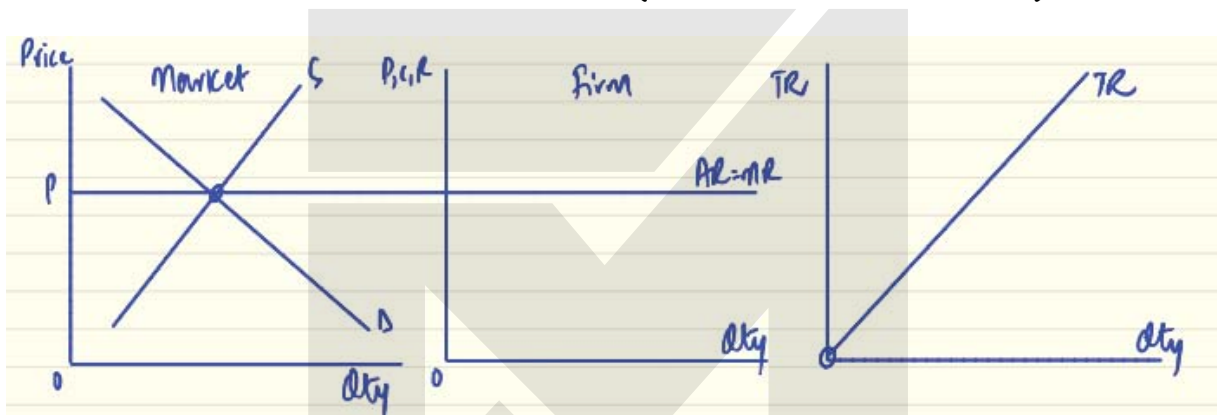
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Theory of demand & supply

$$TR = P \times Q$$

AVERAGE REVENUE: It is the amount of money a firm earns per unit of output sold. It is calculated by dividing the total revenue by the total output sold. $AR = TR/Q$. average revenue is equal to price.

MARGINAL REVENUE: It is the change in the TR brought about by selling one more or less unit of output. $MR = \text{CHANGE IN TR} / \text{CHANGE IN Q}$

REVENUE WHEN PRICE IS CONSTANT (PERFECT COMPETITION)

When the producer sells their output at a constant price, the AR received by the producer will be constant at that price and the TR will rise by that constant price. Under perfectly competitive market structure, an individual firm is a very small part of the total industry and all the firms sell identical products. In this situation an individual firm has no power to affect the market price by altering its own output. In this case the demand curve of the firm's output will be a horizontal line and the AR and MR will be the same. The horizontal demand curve shows that the firm has to sell its output at the prevailing market price which is determined by the market demand and supply curve.

REVENUES WHEN PRICE CHANGES WITH DIFFERENT QUANTITIES SOLD: (IMPERFECT COMPETITION)

In market structures other than perfect competition, the firms face a downward-sloping demand curve. It means when the firm wants to sell more, they have to lower the price. In this case the average revenue, which is the price, falls as output rises. When the demand curve is downward-sloping the MR will be less

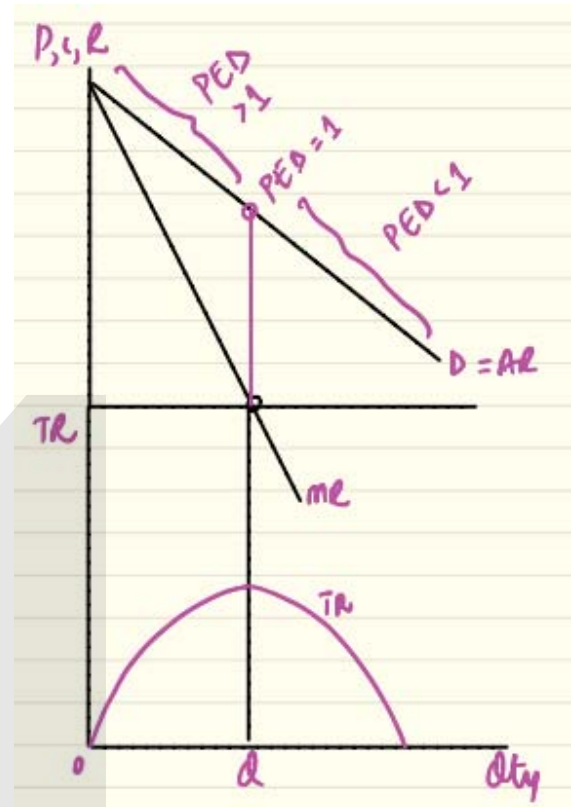
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Theory of demand & supply

than the average revenue and may even become negative. When demand is elastic a fall in price leads to more than proportionate increase in quantity demanded causing TR to rise and MR will be positive, as demand becomes lesser and lesser elastic, MR will fall but still be positive. This fall in MR will lead to a lesser and lesser increase in TR, since MR is the addition to TR and as this additional value is reducing, TR will rise slowly.

When the demand is unitary elastic, a fall in price brings the same proportionate increase in the quantity demanded. There is neither a fall nor a rise in TR i.e. change in TR is zero. In this situation there is no addition to the total revenue i.e. MR is zero. At this point the TR reaches its maximum value. When demand is inelastic, a fall in price will lead to a less than proportionate increase in quantity demanded causing TR to fall and MR will be negative.

The definition of profits: The difference between the revenue the firm gains from selling its output minus the costs of producing that output is called profit. An accountant adds all the monetary costs i.e. the variable costs and the fixed costs and subtracts the costs from total revenue to arrive at profits. An economist also adds the opportunity costs or the implicit costs (e.g. the return on capital employed in alternative businesses) to the monetary costs to calculate the total opportunity costs of the business and then subtracts the costs from revenue to get the pure economic profits.



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